

KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Class: XI

Time: 1.5 hrs.

Day Shift Set-B

Subject: Mathematics (0071)

F.M.: 37

P.M.: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [MCQ] [4×1=4]

Rewrite the correct options of each question in your answer sheet.

1. Which one of the following is logically equivalent to $\sim (\sim p \vee \sim q)$?

- A) $\sim p \wedge q$ B) $\sim p \wedge \sim q$ C) $\sim p \vee \sim q$ D) $p \wedge q$

2. What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+16x^2}}{4+8x}$?

- A) $\frac{1}{2}$ B) 2 C) 4 D) 8

3. Which one of the following is true?

- A) $A \subseteq A \cup B$ B) $A \cap B \subseteq A$ C) $A \cap \bar{A} = \phi$ D) all of the above

4. What is the value of $\lim_{x \rightarrow y} x$?

- A) x B) 0 C) undefined D) y

GROUP - 'B' [5×5=25]

5. a) Define intersections of two sets with an example.

b) Define contradiction of a statement.

c) What is the value of $\lim_{y \rightarrow 0} \frac{y}{\ln(1+y)}$?

d) What the value of $\lim_{x \rightarrow 0} \frac{3^{2x}-1}{x}$?

e) Write the negation of $p \Rightarrow q$.

[1+1+1+1+1=5]

6. Evaluate: a) $\lim_{y \rightarrow 0} \frac{\sqrt{\tan(x+y)} - \sqrt{\tan x}}{y}$

b) $\lim_{x \rightarrow y} \frac{\sqrt{3x-y} - \sqrt{x+y}}{\sqrt{x} - \sqrt{y}}$

[2+3]

7. a) Define set. How does it differ from collection? Justify your answer. [2]

b) For any two non-empty subsets M and N of universal set U , prove that:

i) $\overline{M \cup N} = \bar{M} \cap \bar{N}$

ii) $\overline{M \cap N} = \bar{M} \cup \bar{N}$

[1.5+1.5=3]

8. Evaluate $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$
- Using a table of values.
 - Using algebraic formula.
 - Is it possible to find $f(3)$? Explain. [2+2+1=5]
9. a) Give an example of a function which takes the indeterminate form $\frac{0}{0}$, but whose limit does not exist. [2]
- b) Evaluate: $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x} - \sqrt{x-4})$ [3]
10. a) Define power set with an example. [2]
- b) Construct truth table for the statement $(p \Rightarrow q) \vee (p \wedge \sim q)$ and comment on the result. [3]

GROUP-C [8×1=8]

11. a) Is it possible to write
- $$\lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) \times g(x)]?$$
- Explain with an example. [3]
- b) Under what condition are two statements logically equivalent? Justify your answer. [3]
- c) Does the order of objects in a set matter? Explain. [2]

Handwritten work for question 11a:

$$\frac{\sin(x-a)}{(x-a)} \times \frac{1}{x-a}$$

Best of Luck

$$1 \times \frac{1}{x-a} = \frac{1}{x-a}$$

$$\frac{\sqrt{1+16x^2}}{4+8x} = \frac{\sqrt{1+4x^2}}{2(2+4x)}$$

$$= \frac{\sqrt{1+4x^2}}{4(2+4x)}$$